Quality management system manual

Document no	**RS-QM-01**	**Written by**	Mark.Kim
Rev no	**001**		
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Revised contents	**Drift**	**Review**	**Approval**
	/	/	/

Revised Items	**AS IS**	**TO BE**

Agreement	**QM**	**R&D**					
	Opinion:						

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Phase #I
[Revision history]

Revision history summary			RS-QM01	
			Revision	001
			Confidential	G1
Rev. no	**Rev. date**	**Simple description of changed contents**	**Author**	**Post**
001	2022.09.14	Newly established.		

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Phase #II
[Index]

Phase	Title
I	Revision history
II	Index
III	Company organization
IV	Applied scope
1	Company overview
2	Quality policy
3	Process approach modeling
4	Ethics policy
5	Roles & responsibilities
6	Process list
7	Business review board
8	Definition of terms

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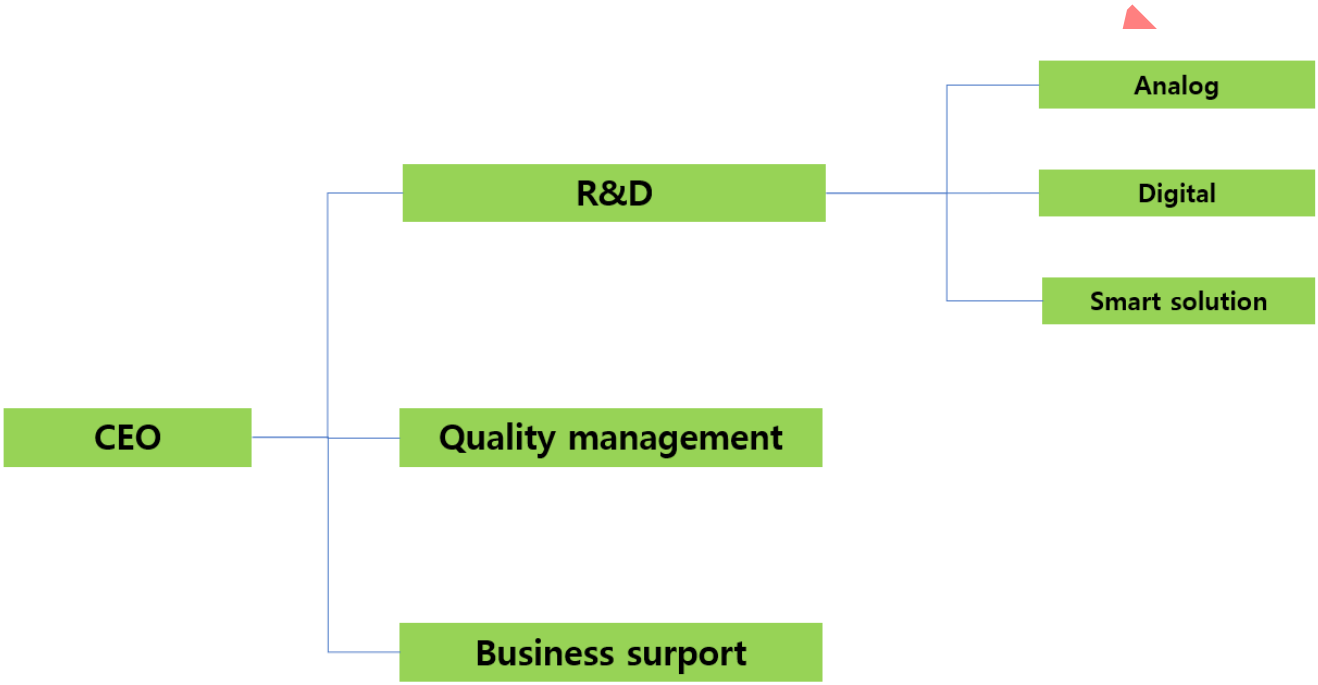
Phase #III
[Company organization]



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Phase #IV
[Applied scope]

1. Scope

This QMS standard specifies requirements for a quality management system where an organization.

a) need to demonstrate its ability to consistently provide products and services that meet customer and applicable statutory and regulatory requirements

b) aims to enhance customer satisfaction through the effective application of the system, including processes for improvement of the system and the assurance of conformity to customer and applicable statutory and regulatory requirements.

This quality manual was prepared based on the CEO's quality policy and the IATF 16949:2016 standard. It is applied to all tasks related to development, production and additional services for automobile parts.

2. Reference standard

ISO 9001:2015 Quality management system

3. Process owner

The person in charge of each process is the management department specified in the process list, and after review by the department head, approval is obtained from the CEO.

Phase #1
[Company overview]

1. Company

ram-sChip.co.Ltd

2. Address

1F, 128ho, changup-ro 42, Sujung-gu,Seongnam-si, Gyeonggi-do, Korea, 13449

3. Product

Interface IP, Automotive IC.

Phase #2
[Quality policy]

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RAMSCHIP QUALITY POLICY

RAMSCHIP.Inc develops market-leading innovative products based on an efficient quality management system, and maximizes our profits by stable supplying market(customer) oriented products with reliability through continuous improvement and securing productivity. And maximize customer & shareholder value.

- RAMSCHIP.Inc operates a quality management system with customer satisfaction as top priority
- RAMSCHIP.Inc ensure the functionality and reliability toward customer oriented products.
- RAMSCHIP.Inc supply defect-free products to the market through quality innovation activities.

Sep.01.2022

ram-sCHIP CEO J.K. Ko.

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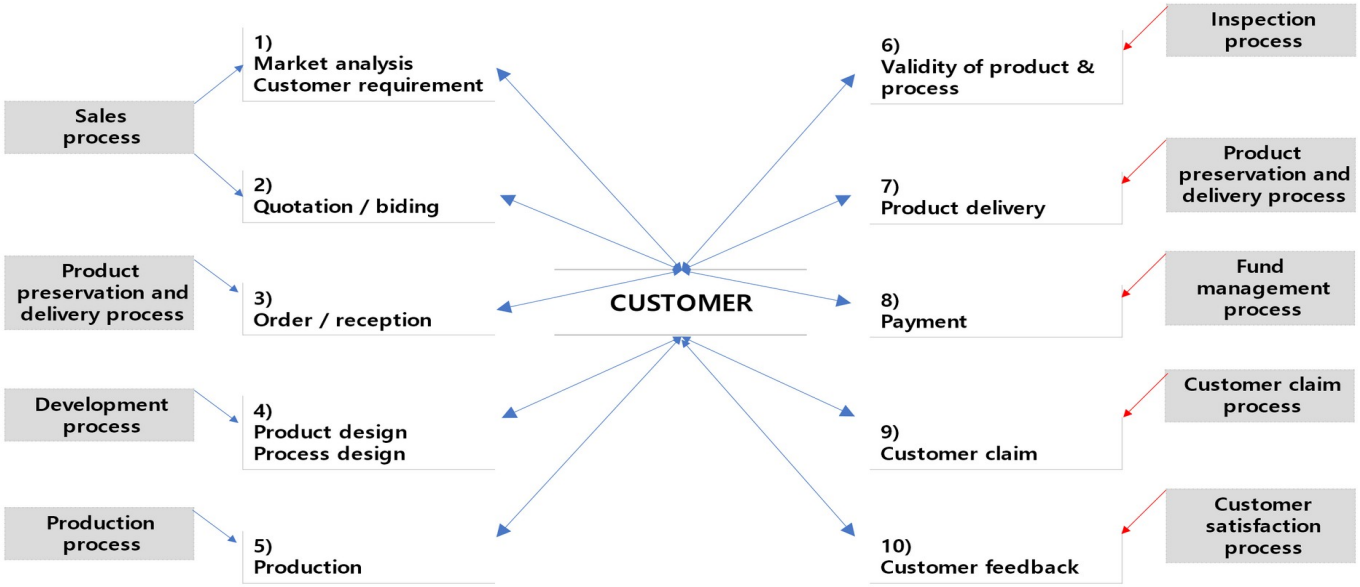
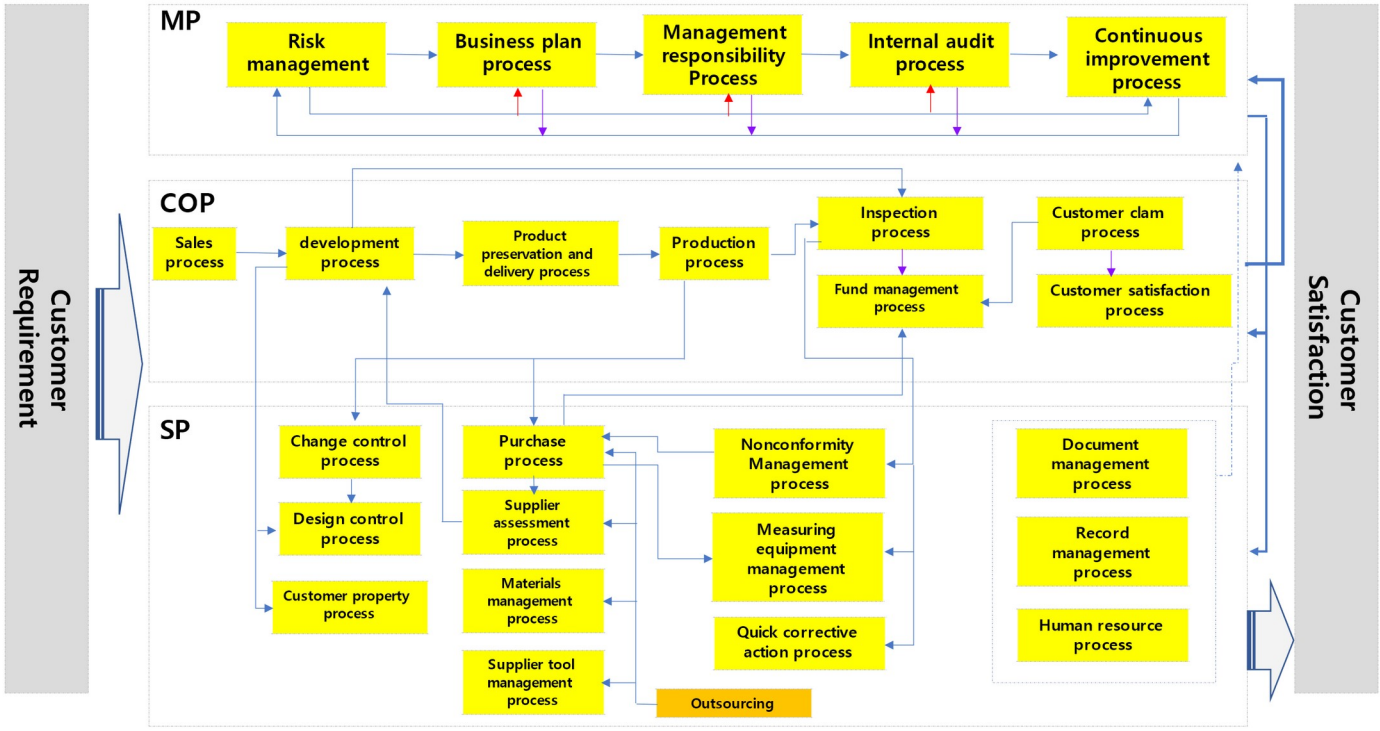
[Process approach modeling]

3-1. Interrelationship between IATF COP and ram-sCHIP COP



3-2. Process map of the ram-sCHIP quality management system.



3-3. Interrelationship matrix between each process.

Classification		Support process (SP)							
		Document Management **RS-SP-01**	Record Management **RS-SP-02**	Design Management **RS-SP-03**	HR Management **RS-SP-04**	Change Control **RS-SP-05**	Purchase Management **RS-SP-06**	Supplier Assessment **RS-SP-07**	Customer Property management **RS-SP-08**
Customer oriented process (COP)	Sales **RS-COP-01**	•	•	•	○	○	○	•	•
	Development **RS-COP-02**	•	•	•	○	•	○	○	•
	Production **RS-COP-03**	•	•	○	○	○	○	○	•
	Inspection **RS-COP-04**	•	•	•	•	•	○	○	•
	Product preservation & delivery **RS-COP-05**	•	•	○	○	○	•	○	○
	Customer satisfaction **RS-COP-06**	•	•	○	○	○	•	•	•
	Customer complaint **RS-COP-07**	•	•	○	○	•	•	○	○
	Fund management **RS-COP-08**	○	○	○	○	○	•	•	○

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Classification		Support process (SP)					Management process (MP)				
		Material management RS-SP-09	Measuring equipment management RS-SP-10	Nonconformity Management RS-SP-11	Supplier tool management RS-SP-11	Quick corrective action RS-SP-12	Risk management RS-MP-01	Management responsibility RS-MP-02	Business plan RS-MP-03	Internal audit RS-MP-04	Continuous improvement RS-MP-05
Customer oriented process (COP)	Sales **RS-COP-01**	•	○	○	○	○	○	○	•	•	•
	Development **RS-COP-02**	○	○	○	○	○	•	○	•	•	•
	Production **RS-COP-03**	•	•	○	○	•	•	○	•	•	•
	Inspection **RS-COP-04**	○	○	•	•	○	•	○	•	•	•
	Product preservation & delivery **RS-COP-05**	○	•	○	○	○	○	○	•	•	•
	Customer satisfaction **RS-COP-06**	•	○	○	•	○	○	○	•	•	•
	Customer complaint **RS-COP-07**	•	○	○	•	○	○	○	•	•	•
	Fund management **RS-COP-08**	•	○	○	○	○	•	○	•	•	•

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3-4. IATF16949:2016 Requirement and related process.

IATF16949:2016	Process number	Process
4. Context of the organization	-	-
4.1. Understanding the organization and its context.	RS-MP-01	Risk management process
4.2. Understanding the needs and expectations of interested parties.	RS-MP-01	Risk management process
4.3. Determining the scope of the quality management system.	RS-QM-01	Quality management system manual
4.3.1. Determining the scope of the quality management system. - supplemental	RS-QM-01	Quality management system manual
4.3.2. Customer specific requirements.	RS-QM-01	Quality management system manual
4.4. Quality management system and its processes	RS-QM-01	Quality management system manual
4.4.1.1. Conformance of products and processes.	RS-COP-4	Inspection process
4.4.1.2. Product safety	RS-COP-02 RS-COP-03	Development process Production process
5. Leadership	-	-
5.1. Leadership and commitment	RS-QM-01	Quality management system manual
5.1.1. General	RS-QM-01	Quality management system manual
5.1.1.1. Corporate responsibility	RS-MP-02	Management responsibility process
5.1.1.2. Process effectiveness and efficiency	RS-QM-01	Quality management system manual
5.1.1.3. Process owner	RS-QM-01	Quality management system manual
5.2. Policy	RS-QM-01	Quality management system manual
5.2.1. Establishing the quality policy	RS-QM-01 RS-MP03	Quality management system manual Business planning process
5.2.2. Communicating the quality policy	RS-QM-01 RS-MP-03	Quality management system manual Business planning process
5.3. Organizational roles, responsibilities and authorities.	RS-QM-01	Quality management system manual
5.3.1. Organizational roles, responsibilities and authorities. - supplemental	RS-QM-01	Quality management system manual
5.3.2. Responsibility and authority for product requirements and corrective actions	RS-QM-01	Quality management system manual
6. Planning	RS-MP-01	Risk management process
6.1. Actions to address risks and opportunities	RS-MP-01	Risk management process
6.1.2.1. Risk analysis	RS-MP-01	Risk management process
6.1.2.2. Preventive action	RS-MP-01	Risk management process
6.1.2.3. Contingency plans	-	Emergency instruction
6.2. Quality objectives and planning to achieve them	RS-MP-03	Business planning process
6.2.2.1. Quality objectives and planning to achieve them – supplement	RS-MP-03	Business planning process
6.3 Planning of changes	RS-QM-01	Quality management system manual

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7. Support	RS-QM-01	Quality management system manual
7.1 Resources	RS-QM-01	Quality management system manual
7.1.1. General	RS-QM-01	Quality management system manual
7.1.2. People	RS-SP-04	HR management process
7.1.3. Infrastructure	RS-COP-03	Production process
7.1.3.1. Plant, facility and equipment planning	RS-COP-02	Development process
7.1.4. Environment for the operation of processes	RS-COP-03	Production process
7.1.4.1. Environment for the operation of processes – supplement	RS-COP-03	Production process
7.1.5. Monitoring and measuring resources	RS-SP-10	Measuring equipment management process
7.1.5.1. General	RS-SP-10	Measuring equipment management process
7.1.5.1.1. Measurement system analysis	RS-SP-10	Measuring equipment management process
7.1.5.2. Measurement traceability	RS-SP-10	Measuring equipment management process
7.1.5.2.1. Calibration/Verification records	RS-SP-10	Measuring equipment management process
7.1.5.3. Laboratory requirements	-	-
7.1.5.3.1. Internal laboratory	-	-
7.1.5.3.2. External laboratory	-	-

IATF16949:2016	**Process number**	**Process**
7.1.6. Organizational knowledge	RS-QM-01	Quality management system manual
7.2. Competence	RS-SP-04	HR management process
7.2.1. Competence - supplemental	RS-SP-04	HR management process
7.2.2. Competence – on-the-job training	RS-SP-04	HR management process
7.2.3. internal auditor competency	RS-SP-04	HR management process
7.2.4. Second party auditor competency	RS-SP-04	HR management process
7.3 Awareness	RS-QM-01	Quality management system manual
7.3.1. Awareness – supplemental	RS-QM-01	Quality management system manual
7.3.2. Employee motivation and empowerment	RS-SP-04	HR management process
7.4. Communication	RS-QM-01	Quality management system manual
7.5. Documented information	RS-SP-01	Document management process
7.5.1. General	RS-QM-01	Quality management system manual
7.5.1.1. Quality management system documentation	RS-SP-01	Document management process
7.5.2. Creating and updating	RS-SP-01	Document management process

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7.5.3. Control of documented information	RS-SP-02	Record management process
7.5.3.2.1. Record retention	RS-SP-02	Record management process
7.5.3.2.2. Engineering specifications	RS-SP-03 RS-SP-05	Design management process Change control process
8. Operation	-	-
8.1 Operation planning and control	RS-COP-01	Sales process
8.1.1. Operation planning and control – supplemental	RS-COP-02	Development process
8.1.2. Confidentiality	RS-COP-02 RS-SP-01	Development process Document management process
8.2 Requirements for products and services	RS-COP-02	Development process
8.2.1. Customer communication	RS-COP-02	Development process
8.2.1.1 Customer communication – supplemental	RS-COP-02	Development process
8.2.2. Determining the requirements for products and services	RS-COP-02	Development process
8.2.2.1. Determining the requirements for products and services - supplemental	RS-COP-02 RS-COP-01	Development process Sales process
8.2.3. Review of the requirements for products and services	RS-COP-02	Development process
8.2.3.1.1. Review of the requirements for products and services - supplemental	RS-COP-02	Development process
8.2.3.1.2. Customer-designated special characteristics	RS-COP-02	Development process
8.2.3.1.3. Organization manufacturing feasibility	RS-COP-02	Development process
8.2.4. Changes to requirements for product and services	RS-COP-02	Development process
8.3. Design and development of products and services	-	-
8.3.1. General	RS-QM-01	Quality management system manual
8.3.1.1. Design and development of product and services - supplemental	RS-COP-02	Development process
8.3.2. Design and development planning	RS-COP-02	Development process
8.3.2.1. Design and development planning – supplemental	RS-COP-02	Development process
8.3.2.2. Product design skills	RS-COP-02	Development process
8.3.2.3. Development of product with embedded software	RS-COP-02	Development process
8.3.3. Design and development inputs	RS-COP-02	Development process
8.3.3.1. Product design input	RS-COP-02	Development process
8.3.3.2. Manufacturing process design input	RS-COP-02	Development process
8.3.3.3. Special characteristics	RS-COP-02	Development process
8.3.4. Design and development controls	RS-COP-02	Development process
8.3.4.1. Monitoring	RS-COP-02	Development process
8.3.4.2. Design and development validation	RS-COP-02	Development process
8.3.4.3. Prototype program	RS-COP-02	Development process
8.3.4.4. Product approval process	RS-COP-02	Development process

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8.3.5. Design and development outputs	RS-COP-02	Development process

IATF16949:2016	Process number	Process
8.3.5.1. Design and development outputs – supplemental	RS-COP-02	Development process
8.3.5.2. Manufacturing process design output	RS-COP-02	Development process
8.3.6. Design and development changes	RS-SP-05	Change control process
8.3.6.1. Design and development changes – supplemental	RS-SP-05	Change control process
8.4. Control of externally provided processes, products and services	-	-
8.4.1 General	RS-SP-09	Material management process
8.4.1.1. General - supplemental	RS-SP-09 RS-SP-06	Material management process Purchase management process
8.4.1.2. Supplier selection process	RS-SP-07	Supplier assessment process
8.4.1.3. Customer-directed sources (also known as "Directed – Buy")	RS-SP-07	Supplier assessment process
8.4.2. Type and extent of control	RS-SP-07	Supplier assessment process
8.4.2.1. Type and extent of control – supplemental	RS-COP-04	Inspection process
8.4.2.2. Statutory and regulatory requirements	RS-COP-04 RS-SP-06	Inspection process Purchase management process
8.4.2.3. Supplier quality management system development	RS-SP-07	Supplier assessment process
8.4.2.3.1. Automotive product-related software or automotive products	N/C	N/C
8.4.2.4. Supplier monitoring	RS-SP-07	Supplier assessment process
8.4.2.4.1. Second-party audits	RS-SP-07	Supplier assessment process
8.4.2.5. Supplier development	RS-SP-07	Supplier assessment process
8.4.3. Information for external providers	RS-SP-07	Supplier assessment process
8.4.3.1. Information for external providers - supplemental	RS-SP-06	Purchase management process
8.5. Product and service provision	-	-
8.5.1. Control of production and service provision	RS-COP02	Development process
8.5.1.1. Control plan	RS-COP02	Development process
8.5.1.2. Standardized work – operator instructions and visual standards	RS-COP-03	Production process
8.5.1.3. Verification of job set-ups	RS-COP-03	Production process
8.5.1.4. Verification after shutdown	RS-COP-03	Production process
8.5.1.5. Total productive maintenance	RS-COP-03	Production process
8.5.1.6. Management of production tooling and manufacturing, test, inspection	RS-SP-11	Supplier tool management process
8.5.1.7. Production scheduling	RS-COP-03	Production process
8.5.2. Identification and traceability	RS-COP-03-I-01	Traceability instruction
8.5.2.1. Identification and traceability – supplemental	RS-COP-03-I-01	Traceability instruction
8.5.3. Property belonging to customers or external providers.	RS-SP-08	Customer property management

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		process
8.5.4. Preservation	RS-COP-05	Product preservation & delivery process
8.5.4.1. Preservation – supplemental	RS-COP-05	Product preservation & delivery process
8.5.5. Post-delivery activities	RS-COP-05	Product preservation & delivery process
8.5.5.1. Feedback of information from service	RS-COP-07	Customer complaint process
8.5.5.2. Service agreement with customer	RS-COP-07	Customer complaint process
8.5.6. Control of change	RS-SP-05	Change control process
8.5.6.1. Control of change – supplemental	RS-SP-05	Change control process
8.5.6.1.1. Temporary change of process controls	RS-SP-05	Change control process
8.6. Release of products and services	RS-COP-04	Inspection process
8.6.1. Release of products and services – supplemental	RS-COP-04	Inspection process
8.6.2. Layout inspection and functional testing	RS-COP-04	Inspection process
8.6.3. Appearance items	RS-COP-04	Inspection process
8.6.4. Verification and acceptance of conformity of external provided Products and services	RS-COP-04	Inspection process
8.6.5. Statutory and regulatory conformity	RS-COP-04	Inspection process
8.6.6. Acceptance criteria	RS-COP-04	Inspection process
8.7. Control of nonconformity outputs	RS-SP-10	Nonconformity management process
8.7.1.1. Customer authorization for concession	RS-SP-10	Nonconformity management process

IATF16949:2016	Process number	Process
8.7.1.2. Control of nonconforming product – customer specified process	RS-SP-10	Nonconformity management process
8.7.1.3. Control of suspect product	RS-SP-10	Nonconformity management process
8.7.1.4. Control of reworked product	RS-SP-10	Nonconformity management process
8.7.1.5. Control of repaired product	RS-SP-10	Nonconformity management process
8.7.1.6. Customer notification	RS-SP-10	Nonconformity management process
8.7.1.7. Nonconforming product disposition	RS-SP-10	Nonconformity management

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		process
9. Performance evaluation	-	-
9.1 Monitoring, measurement, analysis and evaluation	RS-QM-01	Quality management system manual
9.1.1. General	RS-QM-01	Quality management system manual
9.1.1.1. Monitoring and measurement of manufacturing processes	RS-COP-03	Production process
9.1.1.2. Identification of statistical tools	RS-MP-05	Continuous improvement process
9.1.1.3. Application of statistical concepts	RS-COP-02 RS-MP-05 RS-COP-03	Development process Continuous improvement process Production process
9.1.2. Customer satisfaction	RS-COP-06	Customer satisfaction process
9.1.2.1. Customer satisfaction – supplemental	RS-COP-06	Customer satisfaction process
9.1.3. Analysis and evaluation	RS-COP-06	Customer satisfaction process
9.1.3.1. Prioritization	RS-COP-06	Customer satisfaction process
9.2 Internal audit	RS-MP-04	Internal audit process
9.2.2.1. Internal audit program	RS-MP-04	Internal audit process
9.2.2.2. Quality management system audit	RS-MP-04	Internal audit process
9.2.2.3. Manufacturing process audit	RS-MP-04	Internal audit process
9.2.2.4. Product audit	RS-MP-04	Internal audit process
9.3. Management review	RS-MP-02	Management responsibility process
9.3.1. General	RS-MP-02	Management responsibility process
9.3.1.1. Management review – supplemental	RS-MP-02	Management responsibility process
9.3.2. Management review inputs	RS-MP-02	Management responsibility process
9.3.2.1. Management review inputs – supplemental	RS-MP-02	Management responsibility process
9.3.3. Management review outputs	RS-MP-02	Management responsibility process
9.3.3.1. Management review outputs - supplemental	RS-MP-02	Management responsibility process
10. Improvement	RS-QM-01	Quality management system manual
10.1 General	RS-QM-01	Quality management system manual
10.2 Nonconformity and corrective action	RS-SP-11	Nonconformity management process
10.2.3. Problem solving	RS-SP-11 RS-SP-12	Nonconformity management Process Quick corrective action process
10.2.4. Error-proofing	RS-SP-11	Nonconformity management process
10.2.5. Warranty management systems	RS-COP--07	Customer complaint process
10.2.6. Customer complaints and field failure test analysis	RS-COP--07	Customer complaint process
10.3 Continual improvement	RS-MP-05	Continuous improvement process

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10.3.1. Continual improvement – supplemental	RS-MP-05	Continuous improvement process

3-5. Process owner

Classification	PROCESS	QA	Admin	R&D	Sales	Production	Purchase
COP (Customer Oriented Process)	Sales process				•		
	Development process			•			
	Production process					•	
	Inspection process	•					
	Product preservation & Delivery process					•	
	Customer satisfaction Process	•					
	Customer complaint Process	•					
	Fund management Process		•				
SP (Support Process)	Document management Process	•					
	Record management Process	•					
	Design management Process			•			
	HR management process	•					
	Change control process	•					
	Purchase management Process						•
	Supplier assessment Process						•
	Customer property Management process				•		
	Material management Process					•	
	Measuring equipment Management process	•					
	Nonconformity management process	•					
	Quick corrective action	•					

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	Process						
MP (Management Process)	Risk management Process		•				
	Management responsibility process	•					
	Business plan process		•				
	Internal audit process	•					
	Continuous improvement process	•					

Phase #4
[Social responsibility]

Social responsibility	
Code of conduct	1) All employees contribute to the company's best interests rather than interests 2) Conduct own job effectively and efficiently 3) Should avoid taking improper personal advantage by utilize personal position 4) Compliance with company policies and legal requirements. 5) Protection and appropriate use of the company's property. 6) Company confidentiality prohibited: 3rd party 7) No discrimination against partners, competitors and employees 8) Protection of human rights
Anti-corruption	1) All executives and employees should not bribe third parties for profit (public institutions/this party): gifts, entertainment….. 2) All executives and employees shall not use their positions to request money or entertainment for cooperation 3) should not illegally use or embezzle the company's money.
Whistle blow	1)

All employees should understand the above information and faithfully fulfill the stipulated responsibilities and authority.

Sep.01.2022

ram-sCHIP CEO J.K. Ko.

Phase #5
[Division of duties]

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Team	Job	Team	Job
Common	-. Development work -. Customer property work -. Internal / External audit work -. Data analysis work -. Risk management / preventive actions -. Proposal work -. Nonconformity / corrective actions	Quality	-. Customer claim handling -. Nonconformity product management -. Abnormal product handling -. Customer audit -. APQP Process management -. Quality data recording -. Measurement system analysis -. Quality data analysis for improvement -. Internal audit -. Management review host -. Risk management & preventive action. -. Continuous improvement -. Process & product monitoring -. Product qualification -. Measurement facility's calibration & verification host -. QMS management -. Document management -. SCM (Supply chain management) -. AVL(Approved vendor list) management -. Supplier assessment
Development	-. Development work -. Drawing & design management -. Change control management		
Production	-. Yearly production planning -. Process optimization work -. Yield up activity -. Production management -. Product shipping management -. Nonconformity management -. Product traceability management -. Product repair history recording		

Team	Job	Team	Job
Purchase	-. AVL(Approved vendor list) management -. Supplier assessment -. Material purchase (if necessary) -. Negotiation works -. SCM (Supply chain management)	Admin	-. HR (Human resource) management -. Accounting work -. Fund management -. Business planning host -. Internal communication -. Organizational culture management -. Risk management & preventive plan -. Time and attendance management -. Company property management -. Management of the proposal system. -. Patent management -. Execution of funds for purchasing items.
Sales	-. Estimate business -. Customer service -. Order work from customer -. Customer property management -. Country of origin management of raw material or parts. -. Sales strategy planning		

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	-. Market situation analysis		

Phase #6
[Management review]

Meeting	Agenda	Freq	Attend	Owner
Development meeting	-. Development feasibility review -. Development goal review [Quality / Cost / Delivery]	Weekly	-. Sales -. Quality -. Purchase -. Production	Development
Monthly Biz review meeting	-. Each KPI review -. Production issue review -. Quality issue review -. Sales issue review -. Development issue review	Monthly	-. All dept	Quality
Biz strategy meeting	-. Business strategy review -. Company risk review	Half year	-. Admin -. Board members	Admin

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